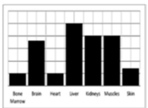
1.1.a. Skeletal and muscular systems

Learners will develop their knowledge and understanding of the roles of the skeletal and muscular systems in the performance of movement skills in physical activities and sport.

Knowledge and understanding of the skeletal system is required and should include the structure and functions of bones, joints and connective tissues.

Knowledge and understanding of planes of movement, the roles of muscles and types of contraction will be developed. Learners will also be able to analyse movement in physical activities and sport applying the underlying knowledge of muscular contraction.

Topic Area	**Content**
Joints, movements and muscles 	• shoulder: ○ flexion, extension, abduction, adduction, horizontal flexion/ extension, medial and lateral rotation, circumduction ○ deltoid, latissimus dorsi, pectoralis major, trapezius, teres minor • elbow: ○ flexion, extension ○ biceps brachii, triceps brachii • wrist: ○ flexion, extension ○ wrist flexors, wrist extensors • hip: ○ flexion, extension, abduction, adduction, medial and lateral rotation ○ iliopsoas, gluteus maximus, medius and minimus, adductor longus, brevis and magnus • knee: ○ flexion, extension ○ hamstring group: biceps femoris, semi-membranosus , semi-tendinosus ○ quadriceps group: rectus femoris, vastus lateralis, vastus intermedius and vastus medialis • ankle: ○ dorsi flexion, plantar flexion ○ tibialis anterior, soleus, gastrocnemius • planes of movement: ○ frontal ○ transverse ○ sagittal.
Functional roles of muscles and types of contraction 	• roles of muscles: ○ agonist ○ antagonist ○ fixator • types of contraction: ○ isotonic ○ concentric ○ eccentric ○ isometric.

Topic Area	Content
Analysis of movement  	• analyse movement with reference to: ○ joint type ○ movement produced ○ agonist and antagonist muscles involved ○ type of muscle contraction taking place.
Skeletal muscle contraction 	• structure and role of motor units in skeletal muscle contraction • nervous stimulation of the motor unit: ○ motor neuron ○ action potential ○ neurotransmitter ○ 'all or none' law.
Muscle contraction during exercise of differing intensities and during recovery 	• muscle fibre types: ○ slow oxidative ○ fast oxidative glycolytic ○ fast glycolytic • recruitment of different fibre types during exercise of differing intensities and during recovery.